

Group Assignment 1

13/05/2026

40 Points Possible

Attempt 1



In Progress

NEXT UP: Submit Assignment

Add Comment

Unlimited Attempts Allowed

22/04/2026

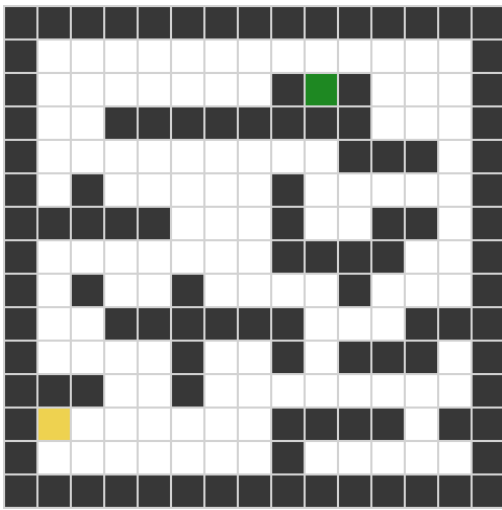
▼ Details

In this assignment, your task is to implement fundamental (tabular) Reinforcement Learning methods, including Dynamic Programming (DP), Monte-Carlo (MC) Methods, and Temporal Difference (TD) Learning, comparing **three algorithms, one from each category**:

1. Policy iteration or Value iteration (from DP)
2. On-policy Monte-Carlo or Off-policy Monte-Carlo (from MC)
3. SARSA or Q-Learning (from TD)

For example, you can choose Value iteration, On-policy Monte-Carlo, and Q-Learning. Any of the 8 above possible combinations is acceptable.

Use the provided simulation environment in the [Project Description](https://canvas.tue.nl/courses/32760/pages/project-description) (<https://canvas.tue.nl/courses/32760/pages/project-description>) to implement your selected algorithms and compute the (optimal) value functions and (optimal) policies across different settings. You need to evaluate your implementations using **at least two different grids** (including "A1_grid.npy", available in the project repository), **at least two values of discount factor** (e.g., 0.6 and 0.9), **at least two values of stochasticity** (e.g., 0.02 and 0.5), in case of MC methods: **at least two values for maximum episode length** (depending on the grid size), in case of TD learning: **at least two values of learning rate** (could be constant or variable), in case of using a soft policy: **at least two values of epsilon** in epsilon-greedy action selection (the use of epsilon decay is also possible, but need to be properly explained in the report), and any other hyperparameter that you think is worth experimenting. Note that the starting position of the robot in all setups must be the same for a fair comparison. In A1_grid.npy, use `agent_start_pos = [1, 12]` as shown below.



Ensure that the problem you address **must be Markov** and you need to find setups where **all three methods reach optimal policies**. This can be achieved by designing a proper environment, state space, and/or reward function. One example of a reward function that can guide the policy to solve the problem could be:

- reward of taking each step: -1 (to encourage the agent to navigate faster and in case of an action that takes the agent to an obstacle, the agent remains in the same tile but gets a penalty for wasting an action)
- reward of reaching a delivery destination: +10

Moreover, consider including a suitable stopping criterion (e.g., if the change in the updates are very small, or number of iterations, etc.).

You need to evaluate the implemented algorithms on as many different experimental setups as your time allows, but at the end, report **6 final setups** to compare all three algorithms and describe how different algorithms work in various settings: *For example, one experiment could assess the effect of the stochasticity parameter on all three different models. In this case, you would keep the other hyperparameters fixed, and use the same grid for all three agents. Make sure to include the cases that some methods work well and some can't find the solution. Explain how different parameters affect learning performance and draw conclusions based on your empirical studies. Additionally, your evaluation should use proper RL metrics rather than problem-specific ones (e.g., total steps or failed moves are **not valid**). Research RL literature for appropriate evaluation metrics. Remember that we expect everyone in the groups to specialize in all parts of the assignment (i.e., being able to implement all three techniques).*

For this assignment, you must submit the following:

1. A **report** that briefly describes the selected algorithms by the groups and how they work, clearly explains at least 6 experimental setups and justify why they are chosen, compares the methods in each setup based on different evaluation metrics and with adequate visualizations to support findings, and draws conclusions based on the obtained results in various configurations.
 - in PDF format which must be submitted separately (for plagiarism check). If the pdf is included in the zip file of the code, points will be deducted.

- **maximum 6 pages**, excluding references (extra pages will be discarded from the evaluation, this also means **no appendix**).
 - must be written in [this LaTeX template](https://www.overleaf.com/read/hqbmhrtspspb)  (<https://www.overleaf.com/read/hqbmhrtspspb>) (without altering the format of the template, you may add/remove/rename sections).
2. A zip file of the **code**, including a clear README file about how to run the code and reproduce the results presented in the report.
- all the codes must be in .py format (we do not accept any other files such as notebook/.ipynb)
 - the code must be **free of errors**, that is, one can execute code using basic steps without facing any problems/errors.
 - if your code requires additional instructions/commands/arguments to be passed in the CLI, this is to be clearly indicated in a guiding README file (.txt/.md will suffice).
 - any additional packages/installations should be clearly indicated in a requirement.txt file.
 - your code must not depend on private files or anything else to which the person running has no access. The code must be able to run from everything provided in the .zip.
 - write your code efficiently so as it won't require hours to train.

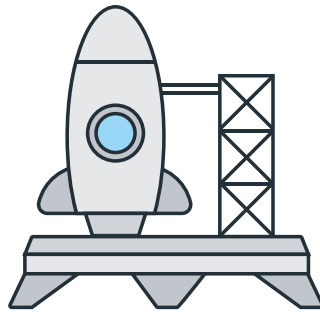
Note that all the files in this assignment need to be **anonymous** for the reviewing assignment: you must remove your names and the group number from the files. Any information/link/etc. that reveals the identity of the group members will negatively impact your grade.

You need to hand in your report+code twice, (i) this assignment and (ii) the [Review of Assignment 1 \(group submission, individual review\)](#)

(<https://canvas.tue.nl/courses/32760/assignments/153549>) (as one zip file). Otherwise, you lose all points of assignment 1 if your work is not submitted on time in FeedbackFruits for reviewing. (both have the same deadline.)

You can consult [this rubric file \(https://canvas.tue.nl/courses/32760/files/6942258?wrap=1\)](https://canvas.tue.nl/courses/32760/files/6942258?wrap=1)  (https://canvas.tue.nl/courses/32760/files/6942258/download?download_frd=1) for preparation of your first assignment (which is also used for reviewing).

Keep in mind, this submission will count for everyone in your Groups2026 group.



Choose a file to upload

File permitted: PDF, ZIP

or

 Canvas Files

☐ I agree to the tool's [End-User License Agreement \(https://canvas-eufrap.int.turnitin.com/integrations/cpf/eula/tue-similarity\)](https://canvas-eufrap.int.turnitin.com/integrations/cpf/eula/tue-similarity).

This assignment submission is my own, original work

*

Submit Assignment